



CUDA-Q Local Deployment Guide

Take it home — deploy locally on your own machine after the tutorial

Installation Guide & Quick Start

Start your CUDA-Q journey

- **Installation by Using PyPI:**

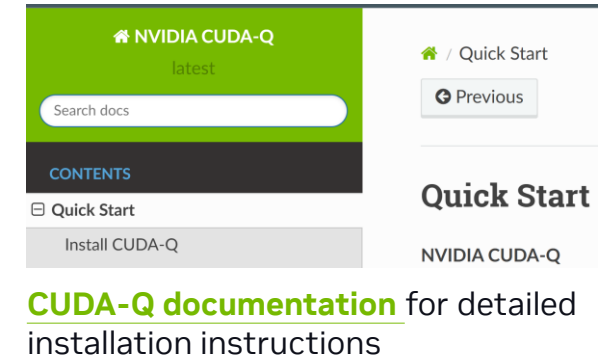
- Required GPU architect: *Turing, Ampere, Ada, Hopper, Blackwell*
- Ensure that the [CUDA Toolkit](#) is installed (version $\geq 12.x$)
(Blackwell supported for CUDA 13.x / Driver 580.65.06+ only)
- [Driver](#) version is updated (version $\geq 560.35.05+$)
- Install CUDA-Q directly [via PyPI](#) (pip version ≥ 24.0)

- **Installation by Using Container [Recommended]:**

- If you are using Windows, WSL (Windows Subsystem for Linux) is needed, and it should be WSL2. Follow [WSL Installation Guide](#)
- Install Docker by visiting [Docker's website](#)
- [Pull/Run the container image](#)
- Container flags can be found on the [NGC page](#), for example:
nvcr.io/nvidia/quantum/cuda-quantum:cu13-0.14.0

- **Get Started:**

- Learn [CUDA-Q basics](#) for easy hands-on
- [Official GitHub repository](#) for full understanding



Open-source Tutorials & CUDA-Q APIs

Let's build your future innovations with CUDA-Q

- **More Tutorials**

- CUDA-Q applications <https://nvidia.github.io/cuda-quantum/latest/using/applications.html>
- Other examples: <https://nvidia.github.io/cuda-quantum/latest/using/examples/examples.html>

- **Advanced Tutorials:**

- [CUDA-Q Academic](#) (recommended)
- [Multi-GPU Workflows](#)

- **All CUDA-Q APIs:**

- https://nvidia.github.io/cuda-quantum/latest/api/languages/python_api

CUDA-Q Applications

Filter by Domain:

All Optimization Chemistry Fundamental Algorithms AI for Quantum Quantum for AI Community

Filter by Backend:

Noiseless Simulator Noisy Simulator QPUs

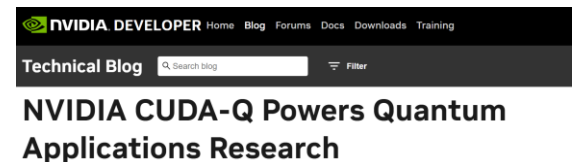
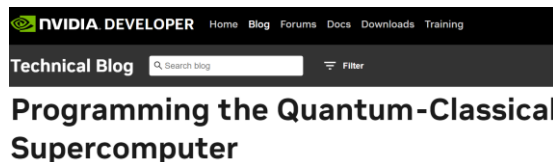
QAOA for Max Cut Problem

Learn the theory behind the Quantum Approximate Optimization Algorithm (QAOA) and how it can be used to solve the Max Cut problem.

#optimization #noiseless #gpu



- **Blogs:**



Example Installation Commands

Install CUDA-Q in Conda environment with pip

- For Windows OS, use WSL terminal to run the following commands and install Miniconda:

```
# Download Miniconda
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh

# Make the installer executable
chmod +x Miniconda3-latest-Linux-x86_64.sh

# Run the installer
./Miniconda3-latest-Linux-x86_64.sh
```

- Create and activate Conda environment, then install CUDA-Q:

```
# Run this command to initialize Conda:
source ~/miniconda3/etc/profile.d/conda.sh

# Check Conda version to verify installation.
conda --version

# Install CUDA-Q with the necessary components for specified CUDA version (e.g., 13.1.0)
cuda_version=13.1.0
conda create -y -n cudaq-env python=3.13 pip
conda install -y -n cudaq-env -c "nvidia/label/cuda-${cuda_version}" cuda
conda env config vars set -n cudaq-env LD_LIBRARY_PATH="$CONDA_PREFIX/envs/cudaq-env/lib:$LD_LIBRARY_PATH"
conda activate cudaq-env
pip install cudaq
```